

Graph Neural Networks for precipitation forecast in state of Rio Grande do Sul

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Section 1

Introduction and objectives

Extreme Climate Event: Rio Grande do Sul - May, 2024



Figure: Gremio's Stadium neighborhood.

Source: Andre Penner.



Figure: Porto Alegre's Historical Center Region.

Source: Ricardo Stuckert.

Section 2

Graph and Dataset

Graph of INMET stations

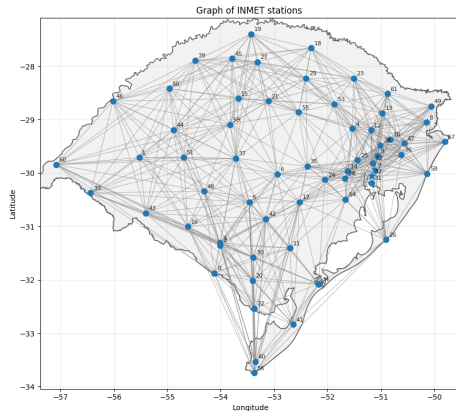


Figure: Graph kNN for $k = 15$.
Source: Lima.

Datasets

- **Data used for training**

- ▶ ERA5 Dataset (ECMWF)
- ▶ Daily meteorological data at locations corresponding to INMET weather stations
- ▶ Dataset period: 01-01-2000 – 31-12-2025

- **Meteorological Variables**

- ▶ Daily precipitation (target variable)
- ▶ Temperature and dew point at sea level (feature)
- ▶ Specific humidity (feature)
- ▶ Vertical velocity at three atmospheric levels (feature)

- **Data Processing**

- ▶ Daily aggregation using mean, maximum, and minimum values
- ▶ Features normalized using z-score normalization
- ▶ Grouping into temporal windows

Source: Copernicus Climate Change Service, Climate Data Store, (2023): ERA5 hourly data on single levels from 1940 to present.

Features per station

- 25 features per day
- Precipitation: 1
- Temperature + dew point: 6 (mean, min, max)
- Specific humidity: 6 (1000 and 850 hPa)
- Vertical velocity: 12 (300, 500 and 850 hPa)

$$x_t^{(i)} \in \mathbb{R}^{25}$$

$$X \in \mathbb{R}^{15 \times N \times 25}$$

$$Y \in \mathbb{R}^{5 \times N}$$

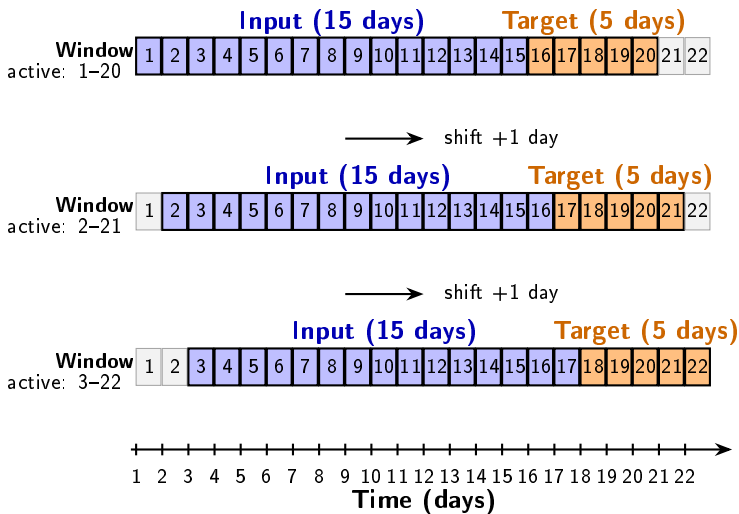


Figure: Sliding Window construction from timeseries data.

Section 3

GNN+LSTM Model

GLSTM architecture

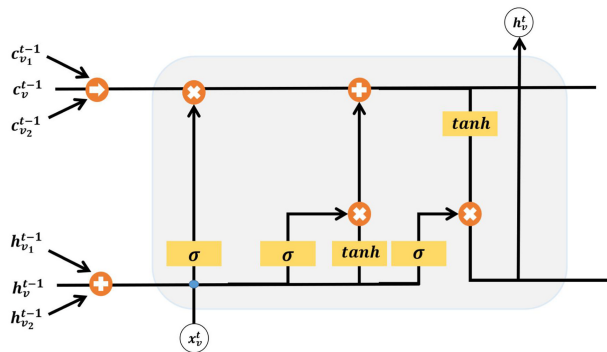


Figure: GLSTM layer architecture.

Source: Gao and Li (2021).

$$\begin{aligned}
 I_t &= \sigma(W^i X_t + U^i H_{t-1} W_{adj} + B^i) \\
 F_t &= \sigma(W^f X_t + U^f H_{t-1} + B^f) \\
 O_t &= \sigma(W^o X_t + U^o H_{t-1} W_{adj} + B^o) \\
 U_t &= \tanh(W^u X_t + U^u H_{t-1} W_{adj} + B^u) \\
 C_t &= I_t \circ U_t + F_t \circ C_{t-1} W_{adj} \\
 H_t &= O_t \circ \tanh(C_t)
 \end{aligned}$$

Section 4

Results

Model Parameters and Loss Function

Model Parameters

- **Graph layer:** Spatial-Convolution
- **LSTM hidden size:** 256
- **LSTM layers:** 2
- **Dropout:** 0.3
- **Optimizer:** AdamW
- **Learning rate:** 10^{-3}
- **Batch size:** 8
- **Weight decay:** 1×10^{-4}

Loss Function

- **Loss function:** quantile mse loss

$$\mathcal{L}_q(y, \hat{y}) = \begin{cases} w_1(y - \hat{y})^2, & y > Q_{0.9}, \\ w_2(y - \hat{y})^2, & y \in [Q_{0.5}, Q_{0.9}], \\ w_{1-q}(y - \hat{y})^2, & y < Q_{0.5} \end{cases}$$

- Asymmetric MSE controlled by quantile q
- Penalizes more the loss in extreme points in the target variable

Results and Discussion

Hidden dim	RMSE	MAE	R^2
256	11.81	8.045	-0.5779
512	12.11	8.46	-0.6608
1024	12.12	10.45	-0.6629

Table: Metrics on test data with all lead days combined.

Prediction analysis



Figure: Prediction for lead day 1.

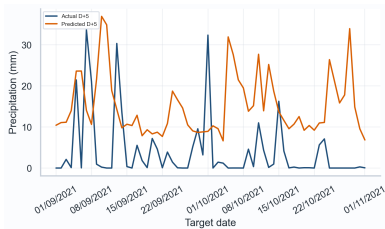


Figure: Prediction for lead day 5.



Figure: Prediction for lead day 1.

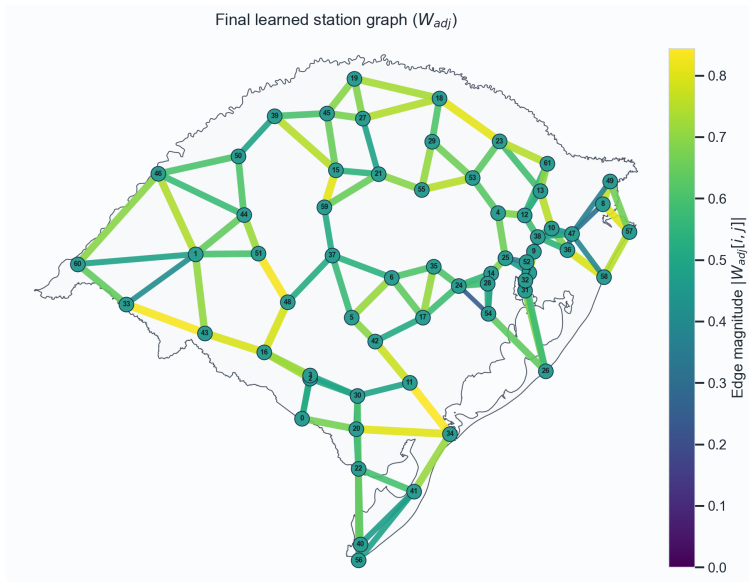


Figure: Prediction for lead day 5.

Animated Prediction vs Real



Learned topology



Future Work

- Improve model performance on extreme precipitation events.
- Evaluate alternative spatial modules, such as spectral graph convolutions (ChebConv and CayleyNet).
- Evaluate the model using observed precipitation data from INMET stations.

Temporary page!

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